Wrist MRI — 30 Sample Cases

A teaching file of 30 worked examples covering the most common wrist MRI diagnoses, with realistic findings and impressions for dictation reference.

TFCC

Case 1 — Traumatic peripheral (foveal) TFCC tear — Palmer 1B

Clinical: 28-year-old male, fall onto outstretched hand during basketball, ulnar-sided wrist pain with painful clicking and DRUJ tenderness.

> **F:** Coronal fat-suppressed PD images show full-thickness disruption of the **peripheral (ulnar) attachment of the TFCC** at the **fovea of the ulnar styloid base**, with loss of the normal low-signal **ligamentum subcruentum** and fluid signal extending to the foveal insertion. The **radial (sigmoid notch) attachment is intact**. Mild fluid distends the **prestyloid recess** and **DRUJ**. Ulnar variance neutral. No lunate marrow edema. Articular cartilage of the ulnar head is preserved. ECU tendon in normal position within its groove.

> **I:** Acute traumatic peripheral (foveal) TFCC tear, Palmer class 1B. Foveal detachment is a potential source of DRUJ instability and is amenable to repair.

Case 2 — Central degenerative TFCC perforation — Palmer 2

Clinical: 54-year-old female, no acute trauma, chronic ulnar-sided wrist ache worse with gripping/twisting.

> **F:** A focal **central perforation of the TFCC articular disc** measures approximately **4 mm**, with fluid signal traversing the disc from the radiocarpal to the DRUJ compartment on coronal PD-FS. Edges are smooth and frayed without peripheral detachment. **Positive ulnar variance of +2 mm.** Mild subchondral marrow edema and a small subchondral cyst in the **proximal ulnar lunate**. Early chondral thinning over the ulnar lunate and ulnar head. Lunotriquetral ligament intact.

> **I:** Degenerative central TFCC perforation (Palmer class 2) in the spectrum of ulnar impaction. Associated positive ulnar variance and early ulnocarpal chondromalacia.

Case 3 — Radial-sided TFCC tear — Palmer 1D

Clinical: 35-year-old male, forced ulnar deviation injury, ulnar wrist pain after a fall while skiing.

> **F:** Coronal images demonstrate detachment of the **TFCC from its radial insertion at the sigmoid notch margin**, with fluid signal undercutting the radial attachment. No associated sigmoid notch avulsion fracture. The **foveal/peripheral attachment is intact**. Ulnar variance neutral. No lunate edema. Small radiocarpal joint effusion.

> **I:** Traumatic radial-sided (sigmoid notch) TFCC tear, Palmer class 1D.

Case 4 — Ulnar impaction syndrome

Clinical: 49-year-old female, chronic ulnar wrist pain, history of prior distal radius fracture.

> **F: Positive ulnar variance of +3 mm.** Subchondral marrow edema with cystic change in the **proximal ulnar aspect of the lunate** and reciprocal changes in the **ulnar head**. Chondral loss over the ulnocarpal articulation. Degenerative central TFCC perforation present. Mild **lunotriquetral ligament** signal/partial tear. No discrete acute fracture.

> **I:** Ulnar impaction (abutment) syndrome with positive ulnar variance, lunate and ulnar head subchondral changes, degenerative TFCC perforation, and LT ligament involvement.

Intrinsic Ligaments

Case 5 — Scapholunate ligament tear with DISI

Clinical: 41-year-old male, FOOSH 6 weeks ago, dorsal wrist pain and weak grip.

> **F:** Complete disruption of the **dorsal band of the scapholunate ligament** with fluid signal through the interval and widening of the **scapholunate interval to 4 mm**. Increased **scapholunate angle of approximately 75°** with dorsal tilt of the lunate consistent with **DISI** alignment. No acute fracture. Mild dorsal capsular edema. TFCC intact.

> **I:** Complete scapholunate ligament tear (dorsal band) with DISI deformity. Predisposes to SLAC-pattern degeneration if untreated.

Case 6 — Lunotriquetral ligament tear with VISI

Clinical: 46-year-old female, ulnar-sided wrist pain after a twisting injury.

> **F:** Disruption of the **lunotriquetral interosseous ligament** with fluid signal across the LT interval. Volar tilt of the lunate with **VISI** alignment pattern. No SL ligament tear. Ulnar variance neutral. Mild ulnocarpal joint fluid. No marrow edema.

> **I:** Lunotriquetral ligament tear with VISI deformity.

Case 7 — Partial scapholunate ligament tear

Clinical: 33-year-old female, climber, gradual dorsoradial wrist pain.

> **F:** Attenuation and intrasubstance increased signal of the **volar and membranous portions of the scapholunate ligament** with an intact **dorsal band**. **Scapholunate interval normal at 2 mm** with no static widening. Normal scapholunate angle (~50°). No DISI. No fracture.

> **I:** Partial (incomplete) scapholunate ligament tear with preserved dorsal band; no static instability.

Carpal Bones & Fractures

Case 8 — Acute scaphoid waist fracture

Clinical: 22-year-old male, FOOSH playing soccer, anatomic snuffbox tenderness, radiographs negative.

> **F:** A nondisplaced low-signal **fracture line through the scaphoid waist** on T1 with surrounding marrow edema on fat-suppressed PD. No cortical step-off or displacement. **Proximal pole marrow signal is preserved** (normal T1 marrow), indicating intact vascularity. No SL ligament tear. Small radiocarpal effusion.

> **I:** Acute nondisplaced scaphoid waist fracture (radiographically occult). Proximal pole vascularity preserved.

Case 9 — Scaphoid proximal pole nonunion with AVN

Clinical: 31-year-old male, remote untreated scaphoid injury, persistent pain and limited motion.

> **F:** Established fracture nonunion at the **scaphoid waist** with sclerotic margins and a fluid-filled cleft. The **proximal pole shows diffuse low T1 signal with low signal on fat-suppressed sequences and no enhancement** on post-contrast images, consistent with **avascular necrosis**. Mild proximal pole fragmentation. Early radioscaphoid chondral changes.

> **I:** Scaphoid waist nonunion with proximal pole AVN (nonviable proximal fragment). Findings relevant for surgical planning (vascularized graft).

Case 10 — Kienböck disease

Clinical: 38-year-old male manual laborer, progressive central wrist pain and stiffness.

> **F:** Diffuse low T1 and heterogeneous fat-suppressed signal throughout the **lunate** with reduced enhancement, in keeping with **avascular necrosis (Kienböck disease)**. Lunate height and contour currently preserved (Lichtman stage II–IIIA). **Negative ulnar variance of −2 mm.** No lunate fragmentation or carpal collapse; carpal height maintained. No SL widening.

> **I:** Kienböck disease (lunate AVN), Lichtman stage II–IIIA, with associated negative ulnar variance.

Case 11 — Hook of hamate fracture

Clinical: 26-year-old male golfer, hypothenar pain after a mishit, point tenderness over the hook.

> **F:** Linear low-signal **fracture through the base of the hook of the hamate (hamulus)** with surrounding marrow edema on fat-suppressed images. Adjacent soft-tissue edema. The **flexor tendons of the small finger** course adjacent without discrete tear. **Ulnar nerve/artery in Guyon canal** are normal. No carpal subluxation.

> **I:** Acute hook of hamate fracture. Watch for ulnar nerve/flexor tendon complications given proximity.

Case 12 — Occult distal radius fracture

Clinical: 67-year-old female, fall on outstretched hand, wrist pain with normal radiographs.

> **F:** Nondisplaced low-signal fracture line through the **distal radial metaphysis** extending toward but not through the articular surface, with extensive surrounding marrow edema on fat-suppressed PD. No intra-articular step-off. No associated TFCC or scapholunate tear. Mild dorsal soft-tissue edema.

> **I:** Radiographically occult nondisplaced distal radius fracture, extra-articular.

Case 13 — Bone contusion (trabecular microtrauma)

Clinical: 19-year-old female, fall while snowboarding, diffuse wrist pain, radiographs negative.

> **F:** Geographic ill-defined marrow edema (low T1, high fat-suppressed PD signal) within the **capitate and dorsal lunate** without a discrete cortical fracture line or cortical breach. No subchondral collapse. Ligaments and TFCC intact. Small joint effusion.

> **I:** Bone contusions (trabecular microtrauma) of the capitate and lunate; no discrete fracture.

Tendons

Case 14 — de Quervain tenosynovitis (first compartment)

Clinical: 32-year-old female, postpartum, radial wrist pain with thumb use, positive Finkelstein.

> **F:** Fluid and synovial thickening surrounding the **abductor pollicis longus and extensor pollicis brevis tendons** within the **first extensor compartment** at the radial styloid, with peritendinous edema and mild thickening of the overlying retinaculum. An **intracompartmental septum** separating APL and EPB is present. Mild tendinosis without tear. Subtle reactive marrow edema in the radial styloid.

> **I:** de Quervain tenosynovitis of the first extensor compartment, with an intracompartmental septum (relevant to treatment response).

Case 15 — ECU tendinosis/tear with subsheath injury and subluxation

Clinical: 29-year-old male tennis player, ulnar wrist pain and snapping with supination.

> **F:** The **extensor carpi ulnaris tendon** is thickened with intrasubstance increased signal and a longitudinal split tear. Surrounding tenosynovial fluid. The **ECU subsheath is disrupted** and the tendon is **subluxed/dislocated medially out of the ulnar groove** on axial images. The ulnar groove appears shallow. Mild reactive edema at the ulnar styloid.

> **I:** ECU tendinosis with longitudinal split tear, subsheath disruption, and tendon subluxation.

Case 16 — Intersection syndrome

Clinical: 40-year-old male rower, dorsoradial forearm pain and swelling proximal to the wrist.

> **F:** Peritendinous edema and fluid surrounding the tendons at the **crossover of the first compartment (APL/EPB) over the second compartment (ECRL/ECRB)**, located approximately **4–5 cm proximal to Lister tubercle** on the dorsoradial distal forearm. Mild tendon thickening without tear. Adjacent muscle edema.

> **I:** Intersection syndrome at the first–second extensor compartment crossover.

Case 17 — Inflammatory (rheumatoid) tenosynovitis

Clinical: 58-year-old female with rheumatoid arthritis, diffuse wrist swelling and pain.

> **F:** Marked synovial thickening and fluid surrounding multiple **extensor tendons (notably the 4th compartment) and the flexor tendons within the carpal tunnel**, with enhancing pannus on post-contrast images. Marginal erosions at the **ulnar styloid and capitate**. No discrete tendon rupture, though the ECU shows attritional thinning. Reactive marrow edema.

> **I:** Inflammatory (rheumatoid) extensor and flexor tenosynovitis with pannus and early marginal erosions; ECU attritional thinning.

Nerves

Case 18 — Carpal tunnel syndrome (median neuropathy)

Clinical: 52-year-old female office worker, nocturnal numbness/tingling in the thumb–middle finger, positive Tinel.

> **F:** The **median nerve is enlarged and flattened** within the carpal tunnel, with increased T2 signal and a **proximal cross-sectional area of approximately 14 mm²** at the pisiform level. **Volar bowing of the flexor retinaculum.** Mild flexor tenosynovitis. Distal flattening of the nerve at the hamate level. No mass or ganglion.

> **I:** Carpal tunnel syndrome with median nerve enlargement, increased signal, and retinacular bowing.

Case 19 — Guyon canal ulnar neuropathy with ganglion

Clinical: 44-year-old male cyclist, ulnar-sided hand numbness and intrinsic weakness.

> **F:** A lobulated **T2-hyperintense ganglion cyst measuring 9 × 6 mm** within **Guyon canal**, arising near the pisotriquetral joint, compressing the **ulnar nerve**, which shows increased T2 signal. Mild edema/early atrophy of the **hypothenar musculature**. Ulnar artery patent. No hook of hamate fracture.

> **I:** Ulnar neuropathy at Guyon canal due to a compressive ganglion cyst, with early hypothenar denervation change.

Other

Case 20 — Dorsal wrist ganglion

Clinical: 27-year-old female, painless dorsal wrist lump, fluctuant on exam.

> **F:** A well-defined unilocular **T2-hyperintense, T1-isointense cystic lesion measuring 12 × 8 mm** along the dorsal wrist, with a thin neck/stalk arising from the **scapholunate interval/dorsal capsule**. No internal enhancement (thin rim only). No solid component. Underlying SL ligament intact.

> **I:** Dorsal wrist ganglion arising from the scapholunate region; benign.

Case 21 — Volar wrist ganglion

Clinical: 36-year-old female, volar radial wrist swelling, mild discomfort.

> **F:** A lobulated **cystic lesion measuring 10 × 7 mm** along the **volar radial wrist** near the radioscaphoid/radiocarpal joint, adjacent to the radial artery, following fluid signal on all sequences with a thin stalk to the joint. No solid enhancing component. Radial artery displaced but patent.

> **I:** Volar wrist ganglion adjacent to the radial artery; benign.

Case 22 — SLAC wrist (scapholunate advanced collapse)

Clinical: 63-year-old male, chronic wrist pain, remote untreated SL injury.

> **F:** Chronic complete **scapholunate ligament tear** with widened SL interval and **DISI**. Stage progression with chondral loss and subchondral sclerosis/cysts beginning at the **radial styloid–scaphoid articulation**, progressing to the **radioscaphoid** and **capitolunate** joints, with proximal capitate migration. The **radiolunate joint is relatively preserved**. Osteophytes and synovitis present.

> **I:** SLAC wrist (scapholunate advanced collapse), stage II–III, with characteristic radioscaphoid and capitolunate involvement and spared radiolunate joint.

Case 23 — SNAC wrist (scaphoid nonunion advanced collapse)

Clinical: 57-year-old male, old scaphoid fracture, progressive wrist pain.

> **F:** Chronic **scaphoid waist nonunion** with degenerative change beginning at the **radial styloid–distal scaphoid fragment**, progressing to the **scaphocapitate and capitolunate** articulations with proximal capitate migration. Subchondral cysts, sclerosis, and osteophytes. Radiolunate joint preserved.

> **I:** SNAC wrist (scaphoid nonunion advanced collapse) with characteristic degenerative pattern sparing the radiolunate joint.

Case 24 — DRUJ instability/injury

Clinical: 34-year-old female, forced rotational injury, painful clicking with forearm rotation.

> **F:** Disruption of the **dorsal and volar radioulnar ligaments (TFCC peripheral attachments)** with foveal detachment. On axial neutral images the **ulnar head is dorsally subluxed relative to the sigmoid notch**. DRUJ effusion. Shallow sigmoid notch morphology. No fracture.

> **I:** DRUJ instability with dorsal subluxation, secondary to peripheral TFCC/radioulnar ligament disruption.

Case 25 — Pisotriquetral osteoarthritis

Clinical: 61-year-old female, focal ulnar/volar wrist pain over the pisiform.

> **F:** Joint space narrowing, subchondral sclerosis, subchondral cysts, and marginal osteophytes at the **pisotriquetral joint** with a small effusion. Mild adjacent FCU peritendinous edema. No fracture. Ulnar nerve in Guyon canal normal.

> **I:** Pisotriquetral osteoarthritis.

Case 26 — Carpal boss

Clinical: 39-year-old male, firm dorsal bump at the base of the index/middle metacarpals, activity-related pain.

> **F:** A bony protuberance with osteophytic spurring at the **second/third carpometacarpal (capitate–trapezoid) dorsal junction** (os styloideum/carpal boss), with associated subchondral cystic change and reactive marrow edema. Overlying adventitial bursal fluid. No fracture.

> **I:** Symptomatic carpal boss at the 2nd/3rd CMC dorsal margin with degenerative change.

Case 27 — Gouty/erosive arthropathy

Clinical: 56-year-old male, recurrent painful wrist swelling, history of hyperuricemia.

F: Lobulated periarticular soft-tissue masses (**tophi**) with intermediate-to-low T1 and heterogeneous T2 signal about the wrist, producing **well-defined erosions with overhanging sclerotic margins** at the carpus and distal ulna. Synovitis and joint effusion. No diffuse osteopenia. Relative joint-space preservation until late.

I: Tophaceous gouty/erosive arthropathy with characteristic juxta-articular erosions and overhanging edges.

Case 28 — Triquetral dorsal avulsion (chip) fracture

Clinical: 30-year-old male, FOOSH with forced wrist flexion, dorsal ulnar wrist pain.

F: A small dorsal cortical **avulsion fragment off the triquetrum** with adjacent marrow and dorsal soft-tissue edema on fat-suppressed images. The remainder of the carpus is intact. Ligaments and TFCC preserved. No carpal malalignment.

I: Dorsal triquetral avulsion (chip) fracture, one of the most common carpal fractures after the scaphoid.

Case 29 — Second extensor compartment tenosynovitis/tendinopathy

Clinical: 45-year-old male weightlifter, dorsoradial wrist pain with resisted extension.

F: Tenosynovial fluid and mild tendinosis of the **second extensor compartment (ECRL/ECRB)** tendons at the level of the radial styloid/Lister tubercle, with peritendinous edema and no discrete tear. Adjacent mild reactive marrow edema. First compartment and intersection region unremarkable.

I: Second extensor compartment tenosynovitis with tendinosis; no tear.

Case 30 — Intraosseous (occult) ganglion

Clinical: 42-year-old female, persistent dorsal wrist pain without a palpable mass.

F: A well-defined **intraosseous ganglion (cyst) measuring 6 mm** in the **proximal scaphoid/lunate** with low T1 and high T2 signal and a thin sclerotic rim, communicating with the overlying capsule via a small stalk. No SL ligament tear and no marrow edema beyond the rim. No occult fracture.

I: Intraosseous (occult) ganglion as a cause of dorsal wrist pain; no ligament tear or fracture.